

ENGR 102 Summer Bridge

Day 11 Student Quiz

While Loops, Update Order, and Termination Evidence

20 points • approximately 20-25 minutes

Name		UIN		Section	
------	--	-----	--	---------	--

Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace a while loop that crosses its target.

5 points

```
current = 2
target = 7
steps = 0

while current < target:
    current = current + 2
    steps = steps + 1
    print(current, steps)
```

Give the complete pass ledger and explain why the final value can exceed target.

--

Question 2. Diagnose a non-progressing loop.

4 points

```
value = 3

while value < 10:
    print(value)
    value + 2
```

Explain why the loop does not terminate, give the minimal repair, and list repaired output.

--

Question 3. Program construction

11 points

Write one complete Python program that satisfies every requirement.

- Read start and target as integers. Valid floors are 1 through 10 inclusive.
- Print invalid if either floor is outside the interval and do not run a loop.
- Print already there if the two valid floors are equal.
- Otherwise use a while loop to move exactly one floor per pass toward target and print each newly reached floor.
- After movement, print Steps: followed by the number of loop passes.

[illegible]

Test evidence / expected result